UL Unit Quiz

CS 7641: Machine Learning

Solutions

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1 Question 1 - Clustering

Part 1. Clustering Scenario (MCMA)

You are analyzing a high-dimensional dataset ($d = 100$) with the following characteristics:

- There are three underlying clusters with very different sizes (one large, two small).
- The data contains moderate Gaussian noise.
- The clusters are roughly spherical in their original feature space.

You are tasked with selecting a clustering method that is robust in high-dimensional, noisy, and imbalanced settings.

Which of the following statements are true in this scenario? **Select all that apply.**

Correct Answers: A, C, F

A. K-Means may be biased toward larger clusters due to its use of Euclidean distance and sum-of-squares optimization.

True. K-Means tends to partition space such that larger clusters can dominate the centroid updates, leading to poor performance on imbalanced datasets.

B. Single linkage clustering is generally preferred in high-dimensional noisy settings due to its chaining property.

False. Single linkage is highly sensitive to noise and often performs poorly in high dimensions due to chaining effects and instability.

C. GMM with EM can handle unequal cluster sizes better than K-Means by learning separate covariance structures and mixing coefficients.

True. EM allows modeling clusters with different sizes and shapes through learned covariance matrices and priors, making it more robust to imbalance.

D. Dimensionality has little effect on clustering because K-Means scales naturally to high dimensions.

False. In high dimensions, distances become less informative (curse of dimensionality), and K-Means performance degrades due to poor separation.

E. Using PCA before clustering will always improve clustering results by preserving local cluster structures.
False. PCA preserves global variance, not necessarily cluster structure, and can sometimes collapse important separations between small or rare clusters.

F. Density-based methods like DBSCAN may fail in this scenario due to varying cluster sizes and the curse of dimensionality.
True. DBSCAN struggles when clusters differ in density and when neighborhood definitions become unreliable in high-dimensional spaces.

Part 2. Segmenting the Condiment Market (MCMA)

Two founders, **Mr. Mayonnaise** and **Dr. Mustard**, are competing in the premium condiments market. They've collected store-level and customer-level features (e.g., weekly unit sales, price, promo spend, shelf space, store size, region, and simple product flags like "organic"/"spicy"). They want to discover actionable market segments using **K-Means** and **GMMs**.

Which statements are true in this context? **Select all that apply.**

A. Standardizing numeric features (e.g., z-scoring sales, price, and promo) is important so no single scale-dominant variable dictates distances.

B. Because GMMs are probabilistic, they inherently handle unscaled features without issues.

C. Adding dozens of one-hot flags for every minor product variant always improves K-Means segmentation.

D. Choosing the number of clusters k by maximizing training likelihood alone guarantees the most useful segments.

E. Increasing k always yields more homogeneous and more actionable segments for marketing.

F. K-Means is sensitive to initialization; multiple restarts or k-means++ help avoid poor local minima.

Part 2. Segmenting the Condiment Market (MCMA) (Answers)

Correct Answers: A, F

A. True. K-Means and GMMs operate in Euclidean feature space; scale-dominant variables (e.g., price vs. shelf width) can overwhelm distances and covariances. Z-scoring numeric features makes contributions comparable, stabilizing both centroid updates (K-Means) and covariance estimation (GMMs).

B. False. GMMs still depend on feature scales: the likelihood and fitted covariances are distorted if one feature has much larger magnitude. Unscaled features bias component shapes and responsibilities. Standardization (and sometimes whitening) remains important.

C. False. Flooding the design with many sparse one-hot flags for minor variants can fragment clusters and inject noise. Prefer parsimonious encodings (coarser taxonomies, target-agnostic embeddings, or dimensionality reduction) aligned with the segmentation goal.

D. False. Maximizing in-sample fit (e.g., likelihood) often over-segments. Use multiple criteria: BIC/AIC for GMMs, silhouette or stability for K-Means, and—critically—downstream business utility (campaignability, lift, coverage).

E. False. Larger k may increase apparent homogeneity but yields tiny, hard-to-activate segments with poor generalization and operational complexity. There is a practical optimum balancing cohesion, separation, and activation cost.

F. True. K-Means can land in poor local minima with bad seeding. Using `k-means++` and multiple restarts improves centroid initialization, stability, and final inertia, especially with noisy retail features.

Part 3. Comparing EM and K-Means in 1D with 3 Clusters (High Variance)

You are given the following 1D dataset of points:

$$\mathcal{D} = \{1, 2, 3, 8, 9, 10, 16, 17, 18\}$$

Assume the data come from **three clusters**. You will compare one iteration of **K-Means** and **EM for Gaussian Mixture Models (GMM)**.

Use the following initialization for both algorithms:

- **K-Means initial centroids:** $\mu_1 = 2$, $\mu_2 = 9$, $\mu_3 = 17$
- **EM initial parameters:**
  - Means: $\mu_1 = 2$, $\mu_2 = 9$, $\mu_3 = 17$
  - Shared, known variance: $\sigma^2 = 9$ (i.e., standard deviation $\sigma = 3$)
  - Mixing coefficients: $\pi_1 = \pi_2 = \pi_3 = \frac{1}{3}$

Now consider a new data point:

$$x_{\text{new}} = 7$$

a) **(K-Means Assignment)** Based on Euclidean distance, which cluster would $x_{\text{new}} = 7$ be assigned to?

b) **(EM Responsibilities)** Compute the posterior probability that $x_{\text{new}} = 7$ belongs to each cluster:

$$\gamma(z = j \mid x = 7) = \frac{\pi_j \cdot \phi(7; \mu_j, \sigma^2 = 9)}{\sum_{k=1}^{3} \pi_k \cdot \phi(7; \mu_k, \sigma^2 = 9)}$$

where

$$\phi(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi \cdot \sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Make sure the responsibilities across all clusters sum to 1. Round each responsibility to 3 decimal places.

Solution: Part 3. Comparing EM and K-Means in 1D with 3 Clusters (High Variance)

Given:

- Cluster means: $\mu_1 = 2$, $\mu_2 = 9$, $\mu_3 = 17$
- Shared variance: $\sigma^2 = 9 \Rightarrow \sigma = 3$
- Equal mixing coefficients: $\pi_1 = \pi_2 = \pi_3 = \frac{1}{3}$
- New point: $x_{\text{new}} = 7$

a) **(K-Means Assignment)**
Compute the Euclidean distance from $x = 7$ to each cluster center:

$$\begin{aligned} d_1 &= |7 - 2| = 5 \\ d_2 &= |7 - 9| = 2 \\ d_3 &= |7 - 17| = 10 \end{aligned}$$

Since the closest center is $\mu_2 = 9$, K-Means assigns $x = 7$ to **Cluster 2**.

b) **(EM Responsibilities)**
Compute the Gaussian probability density:

$$\phi(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi \cdot 9}} \exp\left(-\frac{(x-\mu)^2}{18}\right) \approx \frac{1}{\sqrt{18\pi}} \exp\left(-\frac{(x-\mu)^2}{18}\right)$$

Using:

$$\frac{1}{\sqrt{18\pi}} \approx 0.133$$

Compute unnormalized probabilities:

$$\begin{aligned} \phi_1 &= 0.133 \cdot \exp\left(-\frac{(7-2)^2}{18}\right) = 0.133 \cdot \exp\left(-\frac{25}{18}\right) \approx 0.133 \cdot 0.249 \approx 0.0331 \\ \phi_2 &= 0.133 \cdot \exp\left(-\frac{(7-9)^2}{18}\right) = 0.133 \cdot \exp\left(-\frac{4}{18}\right) \approx 0.133 \cdot 0.801 \approx 0.1066 \\ \phi_3 &= 0.133 \cdot \exp\left(-\frac{(7-17)^2}{18}\right) = 0.133 \cdot \exp\left(-\frac{100}{18}\right) \approx 0.133 \cdot 0.00385 \approx 0.0005 \end{aligned}$$

Multiply by mixing coefficients:

$$\begin{aligned} w_1 &= \frac{1}{3} \cdot 0.0331 \approx 0.0110 \\ w_2 &= \frac{1}{3} \cdot 0.1066 \approx 0.0355 \\ w_3 &= \frac{1}{3} \cdot 0.0005 \approx 0.00017 \end{aligned}$$

Normalize:

$$Z = 0.0110 + 0.0355 + 0.00017 \approx 0.0467$$

Responsibilities:

$$\gamma(z = 1 \mid x = 7) = \frac{0.0110}{0.0467} \approx 0.235$$
$$\gamma(z = 2 \mid x = 7) = \frac{0.0355}{0.0467} \approx 0.760$$
$$\gamma(z = 3 \mid x = 7) = \frac{0.00017}{0.0467} \approx 0.004$$

Conclusion: Under EM, point $x = 7$ has the highest responsibility for Cluster 2 but retains *partial responsibility* for Cluster 1. This demonstrates EM's **soft assignment**, in contrast to K-Means' **hard assignment** to Cluster 2.

2 Question 2 – Feature Selection

Part 1. Challenges in Feature Selection (MCMA)

You are developing a classifier for a biomedical dataset with 500 features and only 120 labeled samples. You aim to improve generalization and interpretability by selecting a subset of features before training.

Which of the following statements are true? **Select all that apply.**

Correct Answers: A, B, D, F

A. High dimensionality with few samples can lead to overfitting if feature selection is not applied.

True. When the number of features greatly exceeds the number of samples, models can fit noise instead of signal, reducing generalization.

B. Wrapper methods may overfit in small datasets because they evaluate performance on the training set.

True. Without proper validation, wrapper methods can select features that appear predictive by chance, especially in low-data regimes.

C. Filter methods are preferred when feature interactions are important to capture.

False. Filters typically rank features independently using metrics like mutual information or correlation, missing important joint effects.

D. Decision tree-based models can serve as embedded methods for feature selection by naturally ranking features based on split criteria.

True. Trees evaluate features during training and can provide importance scores based on how often and how early features are used in splits.

E. Including all features ensures the model captures all possible patterns, especially in small datasets.

False. Including irrelevant or redundant features can increase noise and hurt generalization, especially when data is limited.

F. Cross-validation should be nested when using wrapper methods to avoid selection bias.

True. Nested CV separates the feature selection process from model evaluation, providing an unbiased estimate of generalization performance.

Part 2. Relevance vs. Usefulness in Feature Selection (MCMA)

You are selecting features for a classification model and evaluating both their statistical relevance to the target and their usefulness for improving model performance.

Which of the following statements are true? **Select all that apply.**

Correct Answers: A, B, C, D, E, F

A. A feature can be statistically relevant to the target variable but still be useless in the model if it is redundant with other features.

True. Even highly correlated features can be made redundant by others in the model, diminishing their utility.

B. Filtering methods rank features based on relevance, not usefulness, which may ignore interaction effects.
True. Filter-based techniques typically evaluate features independently and miss synergistic patterns among features.

C. A feature with low correlation to the target might still be useful if it improves model performance when combined with other features.
True. Some features contribute meaningfully only when interacting with others, especially in non-linear models.

D. Wrapping methods can capture both relevance and usefulness by directly evaluating subsets based on model accuracy.
True. Wrappers assess combinations of features in conjunction with a predictive model, incorporating interactions and redundancy.

E. Relevance guarantees usefulness, so highly relevant features should always be included in the final model.
True. In certain controlled settings—such as when using linear models with low feature correlation—strongly relevant features often improve predictive performance. However, this does not imply they should be used without regularization or interaction assessment.

F. A feature may appear irrelevant individually but become useful when combined with others in a non-linear model.
True. This is especially common in tree-based models or neural networks, where features can interact multiplicatively or through complex transformations.

Part 3. Filtering vs. Wrapping (Numerical)

You are working as a data scientist for an agricultural cooperative in the Netherlands. Your team is developing a machine learning model to predict whether a new crop treatment (e.g., fertilizer + irrigation schedule) will lead to a successful yield (1 = success, 0 = failure).

Each field trial contains the following six features:

- **SoilMoisture**: Average moisture content over the season (%)
- **NitrogenLevel**: Initial nitrogen concentration in soil (mg/kg)
- **IrrigationFreq**: Number of irrigation events during growing season
- **AvgTemp**: Average temperature during the season (°C)
- **WeedDensity**: Pre-treatment weed density (plants per m^2)
- **CowsPerAcre**: Number of cows per acre in adjacent pasture

You want to apply feature selection before training a logistic regression classifier.

Filter-Based Method (Pearson Correlation with Yield Outcome) You compute the following absolute correlation values between each feature and crop success:

Feature	Correlation with Yield Success
SoilMoisture	0.70
NitrogenLevel	0.64
IrrigationFreq	0.59
AvgTemp	0.15
WeedDensity	0.05
CowsPerAcre	0.31

You select the top 3 features using this filter method.

Wrapper-Based Method (Forward Selection with Logistic Regression) After evaluating different combinations of features using 5-fold cross-validation, you obtain:

Feature Subset	CV Accuracy
SoilMoisture, NitrogenLevel, IrrigationFreq	0.87
SoilMoisture, IrrigationFreq, CowsPerAcre	0.85
SoilMoisture, NitrogenLevel	0.82
SoilMoisture, IrrigationFreq	0.81
NitrogenLevel, IrrigationFreq	0.75
SoilMoisture, CowsPerAcre	0.78

Questions

a) Based on the filtering approach, which 3 features are selected?

b) Suppose you mistakenly use `CowsPerAcre` instead of `NitrogenLevel`. According to the wrapper results, how much does the cross-validation accuracy change?

c) What is the marginal gain in accuracy from adding `IrrigationFreq` to the pair `SoilMoisture, NitrogenLevel`?

d) Compute the marginal gain in accuracy from adding `CowsPerAcre` to the subset `SoilMoisture, IrrigationFreq`, based on wrapper results.

Solutions: Question 3 – Feature Selection

Part 3. Filtering vs. Wrapping (Numerical)

a) **Based on the filtering approach, which 3 features are selected?**

From the correlation table:

Feature	Correlation
SoilMoisture	0.70
NitrogenLevel	0.64
IrrigationFreq	0.59

The top 3 features selected by filtering are: `SoilMoisture, NitrogenLevel, IrrigationFreq`.

b) **Suppose you mistakenly use `CowsPerAcre` instead of `NitrogenLevel`. According to the wrapper results, how much does the cross-validation accuracy change?**

- Correct subset: `SoilMoisture, NitrogenLevel, IrrigationFreq` $\rightarrow$ Accuracy $= 0.87$
- Mistaken subset: `SoilMoisture, IrrigationFreq, CowsPerAcre` $\rightarrow$ Accuracy $= 0.85$

The accuracy drops from 0.87 to 0.85, a decrease of **0.02**.

c) **What is the marginal gain in accuracy from adding `IrrigationFreq` to the pair `SoilMoisture, NitrogenLevel`?**

- With: `SoilMoisture, NitrogenLevel, IrrigationFreq` $\rightarrow$ Accuracy $= 0.87$
- Without: `SoilMoisture, NitrogenLevel` $\rightarrow$ Accuracy $= 0.82$

Marginal gain $= 0.87 - 0.82 =$ **0.05**

d) **Compute the marginal gain in accuracy from adding `CowsPerAcre` to the subset `SoilMoisture, IrrigationFreq`, based on wrapper results.**

- With: `SoilMoisture, IrrigationFreq, CowsPerAcre` $\rightarrow$ Accuracy $= 0.85$
- Without: `SoilMoisture, IrrigationFreq` $\rightarrow$ Accuracy $= 0.81$

Marginal gain $= 0.85 - 0.81 =$ **0.04**

This shows that `CowsPerAcre`, despite its low individual correlation (0.31), can contribute positively to model performance in the right context.

3 Question 3 - Feature Transformation

Part 1. PCA vs. Polynomial Features (MCMA)

You are building a regression model to predict house prices based on features such as size, age, number of rooms, and location. You are considering two feature transformation techniques:

- **Principal Component Analysis (PCA)** to reduce dimensionality
- **Polynomial feature expansion** to capture nonlinear relationships

Which of the following statements are true in this scenario? **Select all that apply.**
Correct Answers: B, E

A. PCA always improves model accuracy because it reduces noise in the input data.
False. PCA reduces dimensionality and decorrelates features, but it may discard relevant information if important components are excluded.

B. Polynomial features can help the model capture nonlinear patterns such as diminishing returns on house size.
True. Adding interaction and nonlinear terms (e.g., size^2) allows the model to represent complex relationships like saturation effects.

C. PCA expands the feature space by introducing higher-order combinations of existing features.
False. PCA reduces dimensionality by projecting data onto a lower-dimensional subspace—it does not expand the feature set.

D. Polynomial feature expansion reduces the chance of overfitting by simplifying the feature space.
False. Polynomial expansion increases the dimensionality of the data and can *increase* the risk of overfitting, especially with limited data and high-degree terms.

E. PCA may improve performance if the original features are highly correlated, by removing redundancy.
True. PCA is effective when features are collinear, as it projects the data onto orthogonal axes that capture most variance with fewer dimensions.

F. Both PCA and polynomial expansion preserve the interpretability of the original feature space.
False. Polynomial features are more interpretable since they are built from original features. PCA creates new axes that are linear combinations and harder to interpret.

Part 2. Dimensionality Reduction with PCA and ICA (MCMA)

You are working with gene expression data containing 10,000 genes (features) measured across only 80 patient samples. You plan to use PCA or ICA as a preprocessing step before clustering and visualization.

Which of the following statements correctly describe the behavior or risks of PCA and ICA in this setting? **Select all that apply.**
Correct Answers: B, D, F

A. ICA is always preferred over PCA in high-dimensional biological datasets, since it produces independent components.

False. ICA is not always preferred. It requires more data to reliably estimate independent components and may be unstable when the number of features greatly exceeds the number of samples.

B. PCA may produce components aligned with noise if too many are retained in a high-dimensional setting.

True. With many more features than samples, some directions of apparent variance may reflect sampling noise rather than true structure.

C. PCA guarantees recovery of biologically interpretable factors if the top components explain most variance.

False. High variance does not imply interpretability—top PCA components may capture technical artifacts or batch effects rather than true biological signals.

D. Both PCA and ICA may become unstable when features are highly collinear, as is common in gene expression data.

True. Highly correlated features can cause instability in the component estimates, especially when the data matrix is poorly conditioned.

E. Overfitting is not a concern in PCA since there are no model parameters to train.

False. Even unsupervised methods like PCA can overfit by modeling noise as signal when the number of components is too high relative to the number of samples.

F. ICA may identify spurious components if the true latent sources are not sufficiently non-Gaussian.

True. ICA assumes non-Gaussianity to recover independent sources. When this assumption is violated, it can overfit or produce meaningless components.

Part 3. Comparing PCA and ICA in a 7-Sensor Eel Hatchery (Numerical)

You manage a commercial eel hatchery with monitoring systems deployed across multiple tanks. For each tank, sensors continuously measure the following 7 environmental features:

- **Water Temperature** (°C)
- **Dissolved Oxygen** (mg/L)
- **pH Level**
- **Ammonia Concentration** (ppm)
- **Salinity** (ppt)
- **Light Exposure** (lux)
- **Turbidity** (NTU)

You collect **1,000 time points** per tank, standardized and aggregated into a dataset $\mathbf{X} \in R^{1000\times 7}$.

Unknown to you, the hatchery system is governed by 5 latent processes (e.g., filtration control, feeding schedules, etc.), but only 3 dimensions strongly affect variability in the observed data.

You apply both PCA and ICA before clustering the tanks into 6 categories based on environmental dynamics.

PCA Results:

- Eigenvalues of the sample covariance matrix:

$$\lambda = [3.2,\ 2.1,\ 1.0,\ 0.4,\ 0.2,\ 0.05,\ 0.05]$$

- You decide to retain the top 3 principal components for clustering.

ICA Results:

- ICA returns 7 statistically independent components.
- Excess kurtosis values of the components are:

$$\text{kurtosis} = [2.7,\ 1.8,\ 1.1,\ 0.2,\ 0.1,\ -0.1,\ 0.0]$$

- You retain the 3 most non-Gaussian components for clustering.

Questions

a) **Total PCA Variance.** What is the total variance in the data based on the PCA eigenvalues?

b) **Variance Retained.** What percentage of the total variance is retained by the first 3 principal components? Round to the nearest whole number.

c) **ICA Signal Components.** How many ICA components have excess kurtosis above 1?

d) **Non-Gaussianity Sum.** Compute the sum of the three highest excess kurtosis values. What does this suggest about the strength of ICA signal components?

e) **Which of the following best explains why choosing more than 3 components (e.g., 5) might degrade clustering performance?**

A. Including additional ICA components improves completeness but introduces directions with low non-Gaussianity that may correspond to noise.

B. ICA components beyond the top 3 may still be independent but add no clustering signal, reducing silhouette score due to overfitting to noise.

C. PCA and ICA components beyond the third are typically correlated, violating assumptions of the clustering algorithm.

D. Orthogonal components in PCA are better suited for K-Means clustering than the non-orthogonal ICA ones, especially when using more than 3.

Solutions: Part 3. Comparing PCA and ICA in a 7-Sensor Eel Hatchery

a) **Total PCA Variance.**
The total variance is the sum of the eigenvalues:

$$\text{Total variance} = 3.2 + 2.1 + 1.0 + 0.4 + 0.2 + 0.05 + 0.05 = 7.0$$

b) **Variance Retained.**
The top 3 principal components have eigenvalues:

$$3.2 + 2.1 + 1.0 = 6.3$$

Percentage retained:

$$\frac{6.3}{7.0} \times 100 \approx 90\%$$

So, 90% of the variance is retained.

c) **ICA Signal Components.**
We look for ICA components with excess kurtosis greater than 1:

$$[2.7,\ 1.8,\ 1.1] \quad \Rightarrow 3 \text{ components}$$

d) **Non-Gaussianity Sum.**
Sum of the top three excess kurtosis values:

$$2.7 + 1.8 + 1.1 = 5.6$$

This indicates substantial non-Gaussian structure, making these components potentially meaningful for uncovering latent signals.

e) **Component Tradeoff – Dimensionality Selection**

You observe that PCA explains 90% of the variance with the first 3 components, while ICA shows the highest excess kurtosis in its first 3 components. However, ICA also identifies 2 additional components with mild excess kurtosis (0.3 and 0.1). Suppose your downstream task is *unsupervised clustering* of eel behavioral states using these reduced representations.

Which of the following best explains why choosing more than 3 components (e.g., 5) might degrade clustering performance?

A. Including additional ICA components improves completeness but introduces directions with low non-Gaussianity that may correspond to noise.

B. ICA components beyond the top 3 may still be independent but add no clustering signal, reducing silhouette score due to overfitting to noise.

C. PCA and ICA components beyond the third are typically correlated, violating assumptions of the clustering algorithm.

D. Orthogonal components in PCA are better suited for K-Means clustering than the non-orthogonal ICA ones, especially when using more than 3.

Correct Answers: A, B

4 Question 4 - Manifold Learning

Part 1. Manifold Learning (MCMA)

You are handed a 32×32 grayscale dataset of handwritten "A"s where variation is driven primarily by two latent factors: *angle* and *size*. You want a 2D embedding that reveals this curved 2D manifold without crowding or tearing. Consider the following statements about **Sammon Mapping**, **Isomap**, **Laplacian Eigenmaps**, **t-SNE**, and **UMAP**.

Which of the following statements are correct? Select all that apply.

A. **Isomap** constructs a k-NN graph, approximates *geodesic* distances via shortest paths on that graph, and then applies classical MDS to those distances.

B. **Sammon Mapping** minimizes an *unweighted* squared error between high- and low-dimensional distances, intentionally emphasizing large pairwise distances to avoid crowding.

C. **t-SNE** preserves *global* pairwise distances and relative cluster areas, making it reliable for density interpretation and global geometry.

D. **Laplacian Eigenmaps** obtains an embedding from the smallest non-trivial eigenvectors of the generalized eigenproblem $Lf = \lambda Df$, thereby preserving local neighborhoods.

E. **UMAP** requires random initialization in unsupervised mode and cannot inductively map new samples into an existing embedding.

F. **Isomap** is immune to short-circuit errors from spurious edges when k is chosen too large, because its geodesic distances are computed exactly from the manifold equation.

Correct Answers: A, D

A. Isomap: geodesics and MDS.
True. Isomap builds a k-NN graph, estimates *geodesic* distances by shortest paths, then runs classical MDS on that distance matrix to produce the embedding. This helps unwrap manifolds like the Swiss roll and the (angle, size) surface for the "A" images.

B. Sammon Mapping is unweighted and emphasizes large distances.
False. Sammon Mapping uses a *weighted* stress (weights $\propto 1/d^*_{ij}$) that *emphasizes small* original distances, aiming to preserve local structure and reduce crowding relative to plain MDS.

C. t-SNE is globally faithful and preserves areas/densities.
False. t-SNE optimizes local neighborhood fidelity (via KL divergence between high- and low-d affinity distributions). It often distorts global distances and cluster areas; blob size is not a reliable proxy for prevalence.

D. Laplacian Eigenmaps via $Lf = \lambda Df$.
True. Laplacian Eigenmaps constructs a graph over data, forms the graph Laplacian L, and uses the smallest non-trivial eigenvectors of $Lf = \lambda Df$ to embed points while preserving local adjacency.

E. UMAP must start random and cannot transform new data.
False. UMAP commonly uses a spectral initialization by default and supports inductive mapping (`transform`) to place new points into an existing embedding.

F. Isomap is immune to short-circuiting when k is large.
False. If k (or ε) is too large, the graph can introduce shortcuts across the manifold, corrupting geodesic estimates and harming the embedding.

Part 2. Manifold Learning (MCMA)

When t-SNE/UMAP Are the Wrong Tool (MCMA)

Your conservation analytics team is building a *production* embedding for **Roseate Spoonbills** (Platalea ajaja) using high-dimensional data: GPS telemetry, accelerometry, plumage indices, foraging-site salinity, prey proxies, colony metadata, and seasonal weather. You need reliable kNN retrieval, interpretable inter-group distances, stable re-runs, and a well-defined out-of-sample mapping for newly streamed tags.

Which statements correctly explain why 2D t-SNE/UMAP are poor choices for this production problem? Select all that apply.

A. Their 2D maps distort global geometry and densities, so quantitative kNN/retrieval assumptions about distances and neighborhood mass do not hold.

B. They exhibit run-to-run stochasticity and fragile hyperparameter sensitivity, undermining stability required for monitoring and reporting.

C. Out-of-sample mapping is not reliably parametric or metric-preserving, making it risky to place new streaming tags consistently into the same space.

D. Axes lack variance-explained semantics; cluster area/spacing is easily misread as prevalence/similarity, encouraging incorrect ecological conclusions.

E. They provide a guaranteed metric-preserving encoder by default, so distances in 2D can be safely used for population-level inference across seasons.

F. They automatically correct batch effects (e.g., tag hardware generations, survey years), ensuring cross-cohort comparability without extra preprocessing.

Answer Key

Correct Answers: A, B, C, D

A. Global geometry/density distortion breaks kNN assumptions.
True. t-SNE/UMAP emphasize local neighborhoods; inter-cluster distances and apparent densities in 2D are not faithful to the original space, making retrieval unreliable.

B. Stochasticity and hyperparameter fragility.
True. Small changes (random seeds, perplexity/$n_neighbors$, learning rates) can alter layouts—problematic for run-to-run comparability.

C. Unreliable out-of-sample mapping.
True. Vanilla t-SNE lacks a native, stable transform; UMAP's transform is not guaranteed metric-preserving. Streaming tags may land inconsistently.

D. No variance semantics on axes; area/spacing misleads.
True. The 2D coordinates are layout artefacts, not principal components; blob size/spacing can be misinterpreted as prevalence/similarity.

E. Guaranteed metric-preserving encoder by default.
False. Neither method guarantees metric preservation suited for quantitative inference in production.

F. Automatic batch-effect correction.
False. Without explicit correction, embeddings can reflect cohort/tag-year artefacts and exaggerate spurious separations.